Assignment -5

Q1) If Rs. 782 be divided into three parts, proportional to 1/2 : 2/3 : 3/4, then the first part is :

→ Given, $\frac{1}{2}x + \frac{2}{3}x + \frac{3}{4}x = 782$

$= \left(\frac{1}{2} + \frac{2}{3} + \frac{3}{4}\right)x = 782$

$= \left(\frac{12+16+18}{24}\right)x = 782$

$= \left(\frac{46}{24}\right)x = 782 \quad = \left(\frac{23}{12}\right)x = 782$

$\therefore x = \frac{782 \times 12}{23} = 408$

$\therefore 1^{st}$ part $= \frac{1}{2}x = \frac{1}{2}(408) = \underline{\underline{204}}$

Q2) A sum of Rs. 427 is to be divided among A, B and C such that 3 times A's share, 4 times B's share and 7 times C's are all equal. The share of C is :

→ Given, $A + B + C = 427$

$3A = 4B = 7C = x$

Then, $B = \frac{3x}{4}$, $C = \frac{3x}{7}$

$A + \frac{3x}{4} + \frac{3x}{7} = 427$

$\frac{28A + 21A + 12A}{28} = 427$

$61A = 427 \times 28$

$\therefore A = \frac{427 \times 28}{61} = \underline{\underline{196}}$ (A)

$B = \frac{3 \times 196}{4} = \underline{\underline{147}}$ (B)

$C = \frac{3 \times 196}{7} = \underline{\underline{84}}$ (C)

Q3) In college, the ratio of number of boys to girls is 8 : 5. If there are 160 girls, the total number of students in college are :

→ Boys to Girls ratio = 8 : 5

No. of Girls = 160

$\therefore 5x = 160$

$\therefore x = \frac{160}{5} = 32$

No. of boys = $8 \times 32 = 256$

$\therefore$ Total students = $256 + 160 = \underline{\underline{416}}$

Q4) The sum of three numbers is 98, If the ratio of first to second 2 : 3 and that of second to third is 5 : 8, then the second number is :

→ A : B = 2 : 3

B : C = 5 : 8

$= \left(\cancel{5} \times \frac{3}{\cancel{5}}\right) : \left(8 \times \frac{3}{5}\right) = 3 : \frac{24}{5}$

$\therefore A : B : C = 2 : 3 : \frac{24}{5} = \frac{2}{1} \times 5 : \frac{3}{1} \times 5 : \frac{24}{5} \times 5$

$= 10 : 15 : 24$

$\therefore B = \left(\cancel{98}^{2} \times \frac{15}{\cancel{49}}\right) = \underline{\underline{30}}$

LCM

Q5) Six years ago, the average age of A, B, C and D were 30 years. When E joined them, the average of all five became 40 years. The age of E is :

→ Sum of ages of A, B, C, D five years ago = $30 * 4 = 120$

$\therefore$ Sum of ages of A, B, C, D now = $120 + 6 * 4 = 144$

Sum of present ages of all 5 = $40 * 5 = 200$

$\therefore$ Present age of E = $200 - 144 = \underline{56}$

(Q6) The ratio of Seema's age to the age of her mother is 5 : 12. The difference in their ages is 21 years. What will be the ratio of Seema's age and her mother's age after three years?

→ Let the ages be $5x$ and $12x$

The difference between ages is 21 years.

$\therefore 12x - 5x = 21$

$7x = 21$

$\therefore x = 3$

Thus, age of Seema $= 5x = 5 \times 3 = 15$

age of mother of Seema $= 12x = 12 \times 3 = 36$

$\therefore$ Ratio required $= \left(\frac{15+3}{36+3}\right) = \frac{18}{39} = \frac{6}{13}$

$= 6 : 13$

(Q7) The average of 11 numbers is 10.9. If the average of first six numbers is 10.5 and that of the last six numbers is 11.4. The middle number is :

→ Given, average of 11 numbers = 10.9

Avg. of first 6 = 10.5

Avg. of last 6 = 11.4

$\therefore$ Sum of 11 numbers $= 11 \times 10.9$

Sum of first 6 numbers $= 6 \times 10.5$

Sum of last 6 numbers $= 6 \times 11.4$

$\therefore$ The middle number $= (6 \times 10.5) + (6 \times 11.4) - (11 \times 10.9)$

$= 63 + 68.4 - 119.9$

$\rightarrow = 11.5$

Q8) The average monthly salary of 20 employees in an organisation is Rs.1500. If manager's salary is added, then the average salary increases by Rs. 100. What is the manager's monthly salary?

→ Avg. salary of 20 employees = 1500

$\therefore \frac{\text{Sum of salary of 20 employees}}{20} = 1500$

$\therefore$ Sum of salary of 20 employees = 30000 — ①

Given that,

$\frac{\text{Sum of salary of 20 employees + manager salary}}{21 \text{ (manager added)}} = 1500 + 100$

~~= 1600~~ = 1600

$= \frac{30000 + \text{manager salary}}{21} = 1600$

$= 30000 + \text{manager salary} = 1600 \times 21$

$= 30000 + \text{manager salary} = 33600$

$= \text{manager salary} = 33600 - 30000$

$\rightarrow = \underline{\underline{3600}}$

Q9) The average price of three items of furniture is Rs. 15000. If their prices are in the ratio 3 : 5 : 7, the price of cheapest item is:

→ Let the prices be, $3x$ $5x$ and $7x$

$\therefore \frac{3x + 5x + 7x}{3} = 15000$

$3x + 5x + 7x = 45000$

$\therefore 15x = 45000$

$\therefore x = 3000$

$\therefore$ Cheapest item $= 3x = 3(3000) = \underline{\underline{9000}}$

Q10) There are 5 boxes in the cargo. The weight of first box is 200kg and the second box is 20% heavier than the third box which is 25% heavier than the first box. The fourth box which weighs 350 kg is 30% lighter than fifth box. Find the difference between average weight of the four heaviest boxes and the four lightest boxes.

→ Weight of first box = 200 kg – [1]

According to question,

Weight of third box = ~~200 + 25~~ 125% of 200

$= \frac{125}{100} \times 200 = 250\text{ kg}$ – [3]

weight of second box = 120% of 250

$= \frac{\cancel{120}^{6}}{\cancel{100}_{5}} \times \cancel{250}^{50} = 300\text{ kg}$ – [2]

Weight of fourth box = 350 kg – [4]

= 70% of x kg

Weight of fifth box = ~~70% of x~~ x kg. – [5]

70% of x = 350

$\frac{70}{100} x = 350$

$x = \frac{\cancel{350}^{5} \times 100}{\cancel{70}} = 500\text{kg}$

Difference between avg. of heaviest & lightest boxes = ?

Average of 4 heaviest boxes $= \frac{500 + 350 + 300 + 250}{4}$

$= \frac{1400}{4} = 350\text{kg}$

Average of 4 lightest boxes $= \frac{200 + 250 + 300 + 350}{4} = \frac{1100}{4}$

$= 275\text{kg}$

∴ Required difference = 350 - 275

= 75 kg